

fn sample_bernoulli_float

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This proof resides in “contrib” because it has not completed the vetting process.

Warning 1 (Code is not constant-time). `sample_bernoulli_float` takes in a boolean `constant_time` parameter to protect against timing attacks on the Bernoulli sampling procedure. However, the current implementation does not guard against other types of timing side-channels that can break differential privacy, e.g., non-constant time code execution due to branching.

PR History

- Pull Request #473

This document proves that the implementation of `sample_bernoulli_float` in `mod.rs` at commit `f5bb719` (outdated¹) satisfies its proof definition.

`sample_bernoulli_float` considers the binary expansion of `prob` into an infinite sequence `a_i`, like so: $\text{prob} = \sum_{i=0}^{\infty} \frac{a_i}{2^{i+1}}$. The algorithm samples $I \sim \text{Geom}(0.5)$ using an internal function `sample_geometric_buffer`, then returns a_I .

0.1 Hoare Triple

Precondition

Compiler-verified

- Variable `prob` must be of type `T`.
- Variable `constant_time` must be of type `bool`.
- Type `T` has trait `Float`. `Float` implies there exists an associated type `T::Bits` (defined in `FloatBits`) that captures the underlying bit representation of `T`.
- Type `T::Bits` has traits `PartialOrd` and `ExactIntCast<usize>`.
- Type `usize` has trait `ExactIntCast<T::Bits>`.

Caller-verified

None

¹See new changes with `git diff f5bb719..c09efd0 rust/src/traits/samplers/bernoulli/mod.rs`

Pseudocode

```
1 # returns a single bit with some probability of success
2 def sample_bernoulli_float(prob: T, constant_time: bool) -> bool:
3     if prob == 1: #
4         return True
5
6     # prepare for sampling first heads index by coin flipping
7     max_coin_flips = usize.exact_int_cast(T.EXPONENT_BIAS) + usize.exact_int_cast(
8         T.MANTISSA_BITS
9     ) #
10
11     # find number of bits to sample, rounding up to nearest byte (smallest sample size)
12     buffer_len = max_coin_flips.inf_div(8) #
13
14     # repeatedly flip fair coin and identify 0-based index of first heads
15     first_heads_index = sample_geometric_buffer( #
16         buffer_len, constant_time
17     )
18
19     # if no events occurred, return early
20     if first_heads_index is None: #
21         return False
22
23     # find number of zeroes in binary rep. of prob
24     leading_zeroes = (
25         T.EXPONENT_BIAS - 1 - prob.raw_exponent()
26     ) #
27
28     # case 1: index into the leading zeroes
29     if first_heads_index < leading_zeroes: #
30         return False
31
32     # case 2: index into implicit bit directly to left of mantissa
33     if first_heads_index == leading_zeroes: #
34         return prob.raw_exponent() != 0
35
36     # case 3: index into out-of-bounds/implicitly-zero bits
37     if first_heads_index > leading_zeroes + T.MANTISSA_BITS: #
38         return False
39
40     # case 4: index into mantissa
41     mask = 1 << (T.MANTISSA_BITS + leading_zeroes - first_heads_index)
42     return (prob.to_bits() & mask) != 0
```

Postcondition

Theorem 0.1. For any setting of the input parameters `prob` of type `T` restricted to $[0, 1]$, and `constant_time` of type `bool`, `sample_bernoulli_float` either

- raises an exception if there is a lack of system entropy,
- returns `out` where `out` is $\top$ with probability `prob`, otherwise $\perp$.

If `constant_time` is set, the implementation’s runtime is constant.

Proof. To show the correctness of `sample_bernoulli` we observe first that the base-2 representation of `prob` is of the form

$$\text{leading_zeroes} \parallel \text{implicit_bit} \parallel \text{mantissa} \parallel \text{trailing_zeroes}$$

and is represented *exactly* as a normal floating-point number. The [IEEE-754 standard](#) represents a normal floating-point number using an exponent E , and a mantissa m , using a base-2 analog of scientific notation.

Definition 0.2 (Floating-Point Number). A (k, ℓ) -bit floating-point number z is represented as

$$z = (-1)^s \cdot (B.M) \cdot (2^E)$$

where

- s is used to represent the *sign* of z
- B is the implicit bit; 1 for normal floating-point numbers and 0 for subnormal floating point numbers
- $M \in \{0, 1\}^k$ is a k -bit string representing the part of the mantissa to the right of the radix point, i.e.,

$$1.M = \sum_{i=1}^k M_i 2^{-i}$$

- $E \in \mathbb{Z}$ represents the *exponent* of z . When ℓ bits are allocated to representing E , then $E \in [-(2^{\ell-1} - 2), 2^{\ell-1}] \cap \mathbb{Z}$. Note that the range of E is $2^\ell - 2$ rather than 2^ℓ as the remaining two numbers are used to represent special floating point values. When $E = -(2^{\ell-1} - 2)$, then the floating point number is considered *subnormal*.

We now use the technique for **arbitrarily biasing a coin in 2 expected tosses** as a building block. Recall that we can represent the probability **prob** as $\text{prob} = \sum_{i=0}^{\infty} \frac{a_i}{2^{i+1}}$ for $a_i \in \{0, 1\}$, where a_i is the zero-indexed i -th significant bit in the binary expansion of **prob**. Then let $I \sim \text{Geom}(0.5)$ and observe that the random variable a_I is an exact Bernoulli sample with probability **prob** since $P(a_I = 1) = \sum_{i=0}^{\infty} P(a_i = 1 | I = i) P(I = i) = \sum_{i=0}^{\infty} a_i \cdot \frac{1}{2^{i+1}} = \text{prob}$. It is therefore sufficient to show that for any (k, ℓ) -bit float $\text{prob} = \sum_{i=0}^{\infty} \frac{a_i}{2^{i+1}}$, **sample_bernoulli** returns the value a_I with $I \sim \text{Geom}(0.5)$.

First, we observe that by line 3, if **prob** = 1.0 then **sample_bernoulli** returns **true** which is correct by definition of a Bernoulli random variable. Otherwise, the variable **max_coin_flips** is computed to be the value $T : \text{EXPONENT_BIAS} + T : \text{MANTISSA_BITS}$ which equals $2^{\ell-1} - 1 + k$ for any (k, ℓ) -bit float. Since **prob** has finite precision, there is some j for which $a_i = 0$ for all $i > j$. For all (k, ℓ) -bit floating-point numbers, $j \leq 2^{\ell-1} - 1 + k$ by definition. Then **sample_bernoulli** calls **sample_geometric_buffer** with a buffer of length $\lceil \frac{\text{max_coin_flips}}{8} \rceil$ bytes (as shown in lines 9 and 12) which returns **None** if and only if $I > 8 \cdot \lceil \frac{2^{\ell-1} - 1 + k}{8} \rceil$, where $I \sim \text{Geom}(0.5)$ (by Theorem 2.1). In this case, since $I > j$ this index appears in the **trailing_zeroes** part of the binary expansion of **prob** and should always return **false**, i.e., $a_I = 0$ for all $I > j$. We can therefore restrict our attention to when **sample_geometric_buffer** returns an index $I \leq \text{max_coin_flips}$ and show that **sample_bernoulli** always returns a_I .

Assuming that **sample_geometric_buffer** returns some $I < j$, **sample_bernoulli** computes the number of leading zeroes in the binary expansion of **prob** to be **leading_zeroes** = $T : \text{EXPONENT_BIAS} - 1 - \text{raw_exponent}(\text{prob})$, where **raw_exponent**(**prob**) is the value stored in the ℓ bits of the exponent. This value is correct by the specification of a (k, ℓ) -bit float. **sample_bernoulli** then matches on the value **first_heads_index** corresponding to $I \sim \text{Geom}(0.5)$ returned by the function **sample_geometric_buffer**:

Case 1 (**first_heads_index** < **leading_zeroes**).

This corresponds to **sample_geometric_buffer** returning a value I such that a_I indexes into the **leading_zeroes** part of the **prob** variable's binary expansion. Therefore, for any $I < \text{leading_zeroes}$, it follows that $a_I = 0$ and we should return **false**. In this case, **sample_bernoulli** returns **false**.

Case 2 (**first_heads_index** == **leading_zeroes**).

This corresponds to **sample_geometric_buffer** returning a value I such that a_I indexes into the **implicit_bit** part of the **prob** variable's binary expansion. When **prob** is a normal floating point value, i.e., $E \neq -(2^{\ell-1} - 2)$ then the implicit bit $a_I = 1$. Otherwise, when **prob** is a subnormal floating point value, i.e., $E = -(2^{\ell-1} - 2)$, the implicit bit $a_I = 0$. Since **raw_exponent**(**prob**) corresponds to the exponent E for any (k, ℓ) -bit floating

point number `prob`, `sample_bernoulli` returns `true` when `raw_exponent(prob) ≠ 0` and `false` otherwise.

Case 3 (`leading_zeroes + T::MANTISSA_BITS < I`). This corresponds to the case where `sample_geometric_buffer` returns a value I where $I > j$, but $I < \text{max_coin_flips}$ and therefore a_I indexes into the trailing zeroes. In this case, `sample_bernoulli` returns `false` since $a_I = 0$ for all bits in the `trailing_zeroes` part of `prob`'s binary expansion.

Case 4 (`leading_zeroes < first_heads_index < leading_zeroes + T::MANTISSA_BITS`).

This corresponds to `sample_geometric_buffer` returning a value I such that a_I indexes into the mantissa part of the `prob` variable's binary expansion. In this case, `sample_bernoulli` left-shifts the value 1 by $(\text{MANTISSA_BITS} + \text{leading_zeroes} - \text{first_heads_index})$ digits, the index into the mantissa corresponding to the digit a_I in the binary representation of `prob`. Since the operation between the left-shifted 1 and the binary representation of `prob` at that position is a bitwise AND, if the bit in question is 1 (matching the left-shifted 1), `sample_bernoulli` will return `true`. Otherwise, `sample_bernoulli` will return `false`.

Therefore, for any value of `prob`, the function `sample_bernoulli` either raises an exception or returns the value `true` with probability exactly `prob`. □